Chapter 13

Bifurcation Theory

Many problems in mathematics, and its applications to theoretical physics, chemistry, and biology, lead to a problem of the form

$$f(\lambda, x) = 0, \tag{13.1}$$

where f is an operator on $\mathbf{R} \times B_1$ into B_2, with B_1, B_2 Banach spaces. For example, (13.1) could represent a system of differential or integral equations, depending on a parameter λ. We are interested in the structure of the solution set; namely, the set

$$f^{-1}(0) = \{(\lambda, x) \in \mathbf{R} \times B_1 : f(\lambda, x) = 0\}. \tag{13.2}$$

In particular, we seek conditions on f in order that we can determine when a solution $(\bar{\lambda}, \bar{x})$ of (13.1) lies on a "curve" of solutions $(\lambda, x(\lambda))$, at least locally; i.e., for $|\lambda - \bar{\lambda}| < \varepsilon$. We may also inquire as to when $(\bar{\lambda}, \bar{x})$ lies on several solution curves, $(\lambda, x_1(\lambda))$, $(\lambda, x_2(\lambda))$, ...

This last question leads naturally to the concept of a bifurcation point. Thus suppose that $\Gamma: (\lambda, x(\lambda))$, is a curve of solutions of (13.1). Let $(\lambda_0, x_0) \equiv (\lambda_0, x(\lambda_0))$ be an interior point on this curve, with the property that every neighborhood of (λ_0, x_0) in $\mathbf{R} \times B_1$ contains solutions of (13.1) which are not on Γ (Figure 13.1). Then (λ_0, x_0) is called a *bifurcation point* with respect to Γ. Solutions of (13.1) near (λ_0, x_0) and not on Γ, are often loosely referred to as "bifurcating solutions," or the "bifurcation set." Note that the definition does not guarantee the existence of a continuous branch of "bifurcating solutions" emanating from (λ_0, x_0).

Several questions are of interest to us:

(i) Given a curve Γ of solutions of (13.1), what conditions guarantee that it contains a bifurcation point? Obviously, if the implicit function theorem is applicable at a point, bifurcation cannot occur there.

(ii) What is the structure of $f^{-1}(0)$ near a bifurcation point? How is this related to the spectrum of the linearized equations?

(iii) If the bifurcation set is, say, a curve Γ', can it be continued in the large? Does "secondary bifurcation" occur; i.e., does Γ' contain bifurcation points? (see Figure 13.1).

J. Smoller, *Shock Waves and Reaction —Diffusion Equations*
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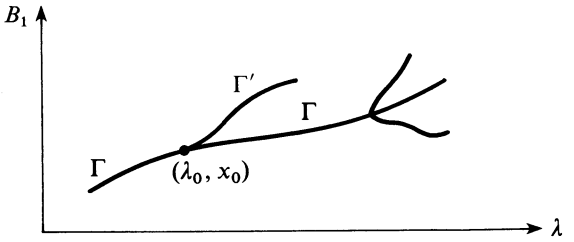


Figure 13.1

(iv) For problems in which solutions of (13.1) are equilibria of a given evolution equation, can one infer any information concerning the stability properties of the bifurcating solution?

We shall attack these problems in this chapter, and somewhat more generally in Chapters 22 and 23. Our results here, for the most part, will come from a rather systematic use of the implicit function theorem. Since our approach will take place in an abstract framework, we shall give a general version of the implicit function theorem. In §A and §B we shall prove some of the standard results in bifurcation theory which stem from the implicit function theorem. In §C we shall prove two general bifurcation theorems using degree theory, while §D is devoted to a description of some different aspects of bifurcation. This is done by giving a fairly complete picture of a special, but interesting, example.

§A. The Implicit Function Theorem

We shall begin by reviewing the definition of the derivative of mappings in finite-dimensional spaces.

Let Ω be an open subset of $\mathbf{R}^m$, and suppose that $f \in C^1(\Omega, \mathbf{R}^n)$. If $a \in \Omega$, then the *derivative* of f at a, written $f'(a)$ or df_a, is the $n \times m$ matrix

$$df_a = \left[\frac{\partial f_i(a)}{\partial x_j}\right],$$

where $f = (f_1, \ldots, f_n)$. Now df_a being a matrix, is thus a bounded linear map from $\mathbf{R}^m$ to $\mathbf{R}^n$; $df_a : x \to df_a x$. Hence $df_a \in B(\mathbf{R}^m, \mathbf{R}^n)$, where we are using the notation $B(X, Y)$ to denote the set of bounded linear maps $X \to Y$.

Since df_x exists for all $x \in \Omega$, we can consider the function $df : \Omega \to B(\mathbf{R}^m, \mathbf{R}^n)$, defined by $x \to df_x$. Observe that $f \in C^1(\Omega, \mathbf{R}^n)$ implies $df \in C(\Omega, B(\mathbf{R}^m, \mathbf{R}^n))$.

If $\xi, a \in \mathbf{R}^m$, and $\phi \in C^1(\Omega, \mathbf{R})$, the *differential* of ϕ at a, acting on ξ is

$$d\phi_a \cdot \xi = \nabla\phi(a) \cdot \xi,$$

where ∇ denotes the usual gradient operator. Applying this last equation to the components of f, we obtain (from Taylor's theorem), the estimate

$$|f(a+\xi) - f(a) - df_a \cdot \xi| = o(|\xi|) \quad \text{as } |\xi| \to 0. \tag{13.3}$$

It is this equation which we shall use to extend the concept of derivative to mappings on Banach spaces.

Thus, let X and Y be Banach spaces, and let $f \in C(\Omega, Y)$, where Ω is open in X.

Definition 13.1. f is (Fréchet) *differentiable* at $a \in \Omega$ if there exists $T \in B(X, Y)$ such that

$$\| f(a+\xi) - f(a) - T\xi \| = o(\| \xi \|) \quad as \ \| \xi \| \to 0.$$

T is called the (Fréchet) *derivative* of f at a, and we write $T = df_a$.

It is not too hard to show that if T exists, then it is unique; see, e.g., [LuS]. If f is Fréchet differentiable at each $a \in \Omega$, we write $f \in C^1(\Omega, Y)$.

Now we want to consider higher derivatives of mappings on Banach spaces. We define the *second derivative* d^2f_a of $f \in C(\Omega, Y)$, at a point $a \in \Omega$, to be the bilinear form $d^2f_a : X \times X \to Y$ which satisfies

$$\| f(a+\xi) - f(a) - df_a\xi - \tfrac{1}{2}d^2f_a(\xi, \xi) \| = o(\| \xi \|^2) \quad \text{as } \| \xi \| \to 0.$$

Continuing in this way, the kth order derivative of f at a is the k-linear mapping $d^kf_a : X^k \to Y$ satisfying

$$\left\| f(a+\xi) - f(a) - df_a\xi - \tfrac{1}{2}d^2f_a(\xi, \xi) - \cdots - \frac{1}{k!}d^kf_a(\xi, \ldots, \xi) \right\|$$
$$= o(\| \xi \|^k) \quad \text{as } \| \xi \| \to 0.$$

If f has a kth order derivative at each $a \in \Omega$, we write $f \in C^k(\Omega, Y)$. It is not very hard to show that $C^k(\Omega, Y)$ is closed under composition; for a proof, see [BP].

It is often useful to have the following theorem; it implies that if $T = I - K$ where K is compact, then dT has the same structure.

Theorem 13.2. *Let $f \in C^1(\Omega, Y)$ be compact near $a \in \Omega$. Then df_a is compact.*

Proof. Suppose the theorem is false. Then there is a $\delta > 0$ and a sequence $\{\xi_i\}$, with $\| \xi_i \| \leq 1$, such that $\| T\xi_i - T\xi_j \| \geq \delta > 0$ for all $i \neq j$. (We are denoting df_a by T.) We can find $\gamma > 0$ so small that whenever $\| \xi \| \leq \gamma$,

$$\| f(a+\xi) - f(a) - T\xi \| < \frac{\delta}{3} \| \xi \|.$$

Now $\|\gamma\xi_i\| \leq \gamma$, so if $i \neq j$,

$$\begin{aligned}\|f(a+\gamma\xi_i) - f(a+\gamma\xi_j)\| &\geq \gamma\|T\xi_i - T\xi_j\| - \|f(a+\gamma\xi_i) - f(a) - \gamma T\xi_i\| \\ &\quad - \|f(a+\gamma\xi_j) - f(a) - \gamma T\xi_j\| \\ &\geq \frac{\delta\gamma}{3}.\end{aligned}$$

Thus, although the sequence $\{a + \gamma\xi_i\}$ is bounded, the sequence $\{f(a + \gamma\xi_i)\}$ obviously has no convergent subsequence. This violates the compactness of f. □

Now in preparation for the implicit function theorem, we consider functions defined on product spaces. Thus, let B_1, B_2 and B_3 be Banach spaces and let U be open in $B_1 \times B_2$. If $f: U \to B_3$, and $u = (u_1, u_2) \in U$, we let U_1 be the cross section, $U_1 = \{x_1 \in B_1 : (x_1, u_2) \in U\}$. We say that f is differentiable with respect to the x_1 variable at (u_1, u_2) if the function $g(x_1) = f(x_1, u_2)$ is differentiable at u_1. When this holds, we write $dg_u = D_1 f(u_1, u_2)$; dg_u is a linear mapping from U_1 into B_3. f is said to be differentiable with respect to x_1 on U, if it is differentiable with respect to x_1 at each $u \in U$. Of course, the usual properties for partial derivatives hold in this general context. In particular, if f is differentiable at $u = (u_1, u_2) \in U$, then f is differentiable with respect to both x_1 and x_2 at u, and for all $(\xi_1, \xi_2) \in B_1 \times B_2$,

$$df_u(\xi_1, \xi_2) = D_1 f(u) \cdot \xi_1 + D_2 f(u) \cdot \xi_2.$$

Furthermore, the mapping $(x_1, x_2) \to (D_1 f(x_1, x_2),\ D_2 f(x_1, x_2))$ is in $C(U, B(U_1, B_3) \times B(U_2, B_3))$. Finally, if $f \in C(U, B_3)$, and f is continuously differentiable with respect to both variables (i.e., $D_1 f: U \to B(U_1, B_3)$ and $D_2 f: U \to B(U_2, B_3)$, are continuous), then $f \in C^1(U, B_3)$. The proofs of these statements are easy; see, e.g., [BP].

We can now state the implicit function theorem in the Banach space context; we omit the proof (see, e.g., [Cr 2]).

Theorem 13.3 (Implicit Function Theorem). *Let $f \in C(U, B)$ where U is open in $\Lambda \times B_1$, and Λ, B_1 and B are Banach spaces. Assume that:*

(i) *$f(\lambda_0, u_0) = 0$ for some $(\lambda_0, u_0) \in U$,*
(ii) *$D_2 f: (\lambda, u) \to D_2 f(\lambda, u)$ is continuous in a neighborhood of (λ_0, u_0), and*
(iii) *$D_2 f(\lambda_0, u_0)$ is nonsingular (i.e., has a bounded inverse); equivalently, $D_2 f(\lambda_0, u_0)$ is a continuous bijective (linear) mapping.*

Then there exists a continuous "curve" $u = u(\lambda)$ defined in a neighborhood N of λ_0, such that $u(\lambda_0) = u_0$, and $f(\lambda, u(\lambda)) \equiv 0$ in N. These are the only solutions of $f(\lambda, u) = 0$ in N. Finally, if $f \in C^k(U, B)$, then $u \in C^k(N, B)$.

Thus, according to the implicit function theorem, we can only expect bifurcation at (λ_0, u_0) if $D_2 f(\lambda_0, u_0)$ is singular. On the other hand, even if $D_2 f(\lambda_0, u_0)$ is singular, (λ_0, u_0) need *not* be a bifurcation point. This of course, is all quite well known. However, let us observe that if $F(\lambda, x) = \lambda - x^3$, then $F(0, 0) = 0 = F_x(0, 0)$; yet the equation $F(\lambda, x) = 0$ uniquely determines x as a function of λ. On the other hand, if $G(\lambda, x) = \lambda - x^2$, then $G(0, 0) = 0 = G_x(0, 0)$ and $(0, 0)$ is a bifurcation point. The difference comes from a consideration of the derivatives, $F_x = -3x^2$ and $G_x = -2x$; namely, the latter changes sign at $x = 0$, while the former does not.

In order to get a better feeling about what is happening here, let's consider the case where f is a mapping on a finite-dimensional space, where for simplicity, we assume that $f(\lambda, 0) = 0$ for all $\lambda \in \mathbf{R}$. Using Taylor's theorem, we can write

$$f(\lambda, u) = L_0 u + (\lambda - \lambda_0) L_1 u + r(\lambda, u), \tag{13.4}$$

where $L_0 = D_2 f(\lambda_0, 0)$ and $L_1 = D_1 D_2 f(\lambda_0, 0)$ are $n \times n$ matrices, and $r \in C^2$ satisfies

$$r(\lambda, 0) \equiv 0, D_2 r(\lambda_0, 0) = D_1 D_2 r(\lambda_0, 0) = 0. \tag{13.5}$$

Now one can show, using degree theory, that if $\det(L_0 + (\lambda - \lambda_0)L_1)$ changes sign at λ_0, then $(\lambda_0, 0)$ is a bifurcation point; (see [Kr, p. 196]). We shall not prove this here since we are going to prove a more general theorem in §C. What we will prove is the so called "bifurcation from a simple eigenvalue" theorem. Here is a preliminary, finite-dimensional statement.

Theorem 13.4. *Let U be an open subset of $\mathbf{R} \times \mathbf{R}^n$ and let $f \in C^2(U, \mathbf{R}^n)$ be given by* (13.4), *where r satisfies* (13.5). *Assume that the null space of L_0 is spanned by u_0, and that $L_1 u_0$ is not in the range of L_0. Denote $\{u_0\}^\perp$ by Z. Then there is a $\delta > 0$ and a C^1-curve $(\lambda, \phi): (-\delta, \delta) \to \mathbf{R} \times Z$ such that* (i) $\lambda(0) = \lambda_0$, (ii) $\phi(0) = 0$, *and* (iii) $f(\lambda(s), s(u_0 + \phi(s))) = 0$ *for* $|s| < \delta$. *Furthermore, there is a neighborhood of $(\lambda_0, 0)$ such that any zero of f either lies on this curve or is of the form $(\lambda, 0)$.*

Thus $(\lambda_0, 0)$ is a bifurcation point for f. This situation is depicted in Figure 13.2. Note that we get some information about the "direction" of bifurcation; i.e., $du/ds = u_0$ at $s = 0$. Furthermore, if we use the notation $N(T)$ and $R(T)$ to denote, respectively, the null space and range of an operator T, then the conditions $N(L_0) = \text{span}\{u_0\}$, and $L_1 u_0 \notin R(L_0)$ are equivalent to the fact that $\mu = 0$ is a simple root of $\det(L_0 + \mu L_1)$; hence, in particular, $\det(L_0 + (\lambda - \lambda_0)L_1)$ changes sign at $\lambda = \lambda_0$.

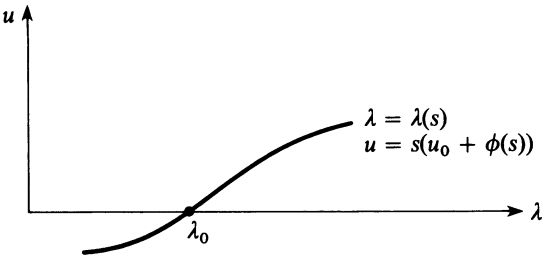


Figure 13.2

Proof of Theorem 13.4. The idea is to introduce a new parameter s which enables us to immediately apply the implicit function theorem on the "quotient" space Z. Thus, we define a new function $F(s, \lambda, z)$ by

$$F(s, \lambda, z) = \begin{cases} s^{-1}f(\lambda, s(u_0 + z)), & \text{if } a \neq 0, \\ D_2 f(\lambda, 0)(u_0 + z), & \text{if } s = 0. \end{cases}$$

Using (13.4) we can easily check that $F \in C^1(U \times Z, \mathbf{R}^n)$. Now since $F(0, \lambda_0, 0) = 0$, and[1] $D_2 D_3 F(0, \lambda_0, 0) \cdot (\tilde{\lambda}, \xi) = L_0 \xi + \tilde{\lambda} L_1 u_0$ is nonsingular, we can apply the implicit function theorem, and solve $F(s, \lambda, z) = 0$, thereby obtaining functions $\lambda = \lambda(s)$, $z = \phi(s)$ defined for $|s| < \delta$, near the point $(0, \lambda_0, 0)$. For the uniqueness, we observe that since $D_2 D_3 F(0, \lambda_0, 0)$ is an isomorphism, it follows by a continuity argument, that for every ε, $0 < \varepsilon < 1$, there is a neighborhood N of $(\lambda_0, 0)$ with the property that every solution $(\lambda, u) \equiv (\lambda, su_0 + w)$ of $f = 0$ (with $w \in Z$) in N satisfies the estimate $\|w\| < 2|s|\varepsilon$, if $|s| + |\lambda - \lambda_0|$ is small. Indeed, if $(\lambda, u) \in U$, $w \in Z$, then since the mapping $(\lambda, w) \to L_0 w + (\lambda - \lambda_0)L_1 u_0$ is nonsingular, there is a $k > 0$ such that

$$L_0 w + (\lambda - \lambda_0)L_1 u_0 = v \quad \text{implies} \quad \|w\| + |\lambda - \lambda_0| < k\|v\|. \tag{13.6}$$

Now given any $\varepsilon > 0$, the hypotheses (13.5) on r together with the fact that $r(\lambda, su_0 + w) - r(\lambda, 0) = r(\lambda, su_0 + w)$, imply that there exists a $\delta > 0$ such that

$$\|w\| + |s| + |\lambda - \lambda_0| < \delta \quad \text{implies} \quad \|r(\lambda, su_0 + w)\| < \varepsilon(|s| + \|w\|). \tag{13.7}$$

But noting that $f(\lambda, su_0 + w) = 0$ can be written in the form

$$L_0 w + s(\lambda - \lambda_0)L_1 u_0 = -(\lambda - \lambda_0)L_1 w + r(\lambda, su_0 + w),$$

[1] Namely, $F(0, \lambda, z) = D_2 f(\lambda, 0)(u_0 + z)$, so $D_3 F(0, \lambda, 0)\xi|_{\lambda=0} = D_2 f(0, 0)\xi = L_0 \xi$, and $D_2 F(0, 0, 0)\tilde{\lambda} = \tilde{\lambda} D_1 D_2 f(0, 0)u_0 = \tilde{\lambda} L_1 u_0$. Thus $d_{(\lambda, z)}F(\xi, \tilde{\lambda}) = L_0 \xi + \tilde{\lambda} L_1 u_0$.

we find that (13.6) and (13.7) yield

$$\|w\| + |s|\,|\lambda - \lambda_0| \leq k(|\lambda - \lambda_0|\,\|L_1\|\,\|w\|) + \varepsilon(|s| + \|w\|).$$

Thus, if $0 < \varepsilon < 1$,

$$\tfrac{1}{2}\|w\| < (1 - k|\lambda - \lambda_0|\,\|L\| - \varepsilon)\|w\| \leq \varepsilon|s|, \quad \text{if } |\lambda - \lambda_0| \text{ is small.}$$

From this estimate it follows that if $u \neq 0$, then $s \neq 0$ (otherwise $su_0 + w = 0$), and so $F(s, \lambda, s^{-1}w) = 0$. Thus, by the uniqueness part of the implicit function theorem, if ε is sufficiently small, $(s, \lambda, s^{-1}w) = (s, \lambda(s), \phi(s))$. Hence $\lambda = \lambda(s)$, and $u = su_0 + w = su_0 + s\phi(s)$. This completes the proof. □

Now let us consider the analogue of this theorem in the infinite-dimensional context. We will often assume the condition

$$f(\lambda, 0) \equiv 0, \qquad \lambda \in S \subset \mathbf{R}, \tag{13.8}$$

where S will vary according to the specific context. Here is the infinite-dimensional version of Theorem 13.4.

Theorem 13.5 (Bifurcation from a Simple Eigenvalue). *Let $U = S \times V$ be an open subset of $\mathbf{R} \times X$ and let $f \in C^2(U, Y)$, where X and Y are Banach spaces. Suppose that* (13.8) *holds, and let $L_0 = D_2 f(\lambda_0, 0)$ and $L_1 = D_1 D_2 f(\lambda_0, 0)$. Assume that the following conditions hold:*

(i) *$N(L_0)$ is one-dimensional, spanned by u_0.*
(ii) *$R(L_0)$ has co-dimension 1; i.e., $\dim[Y/R(L_0)] = 1$.*
(iii) *$L_1 u_0 \notin R(L_0)$.*

Let Z be any closed subspace of X such that $X = [\operatorname{Span} u_0] \oplus Z$ (i.e., any $x \in X$ can be uniquely written as $x = \alpha u_0 + z$, $\alpha \in \mathbf{R}$, $z \in Z$). Then the conclusions of Theorem 13.4 *hold.*

The proof of this theorem is virtually identical to that of Theorem 13.4. We remark that in the finite-dimensional case, $\dim N(L_0) = 1$ is equivalent to $\operatorname{codim} R(L_0) = 1$; this is not necessarily true in infinite dimensions.

We now give two examples to illustrate this theorem.

EXAMPLE 1. Let $[0, 1]$ be the unit interval in $\mathbf{R}$, and let C^k, $k = 0$ or 2 be the Banach space $C^k([0, 1], \mathbf{R})$, with its usual norm. Denote by $C_0^{2+\alpha}$, the subspace $\{u \in C^{2+\alpha}: u'(0) = u'(1) = 0\}$. Let $f: \mathbf{R} \times C_0^{2+\alpha} \to C^\alpha$ be defined by

$$f(\lambda, u) = u'' + \lambda[u + u^2 h(u)],$$

where h is smooth. Then $f(\lambda, 0) = 0$ for all $\lambda \in \mathbf{R}$,

$$D_2 f(\lambda, 0)v = v'' + \lambda v \quad \text{and} \quad D_1 D_2 f(\lambda, 0)v = v, \qquad v \in C_0^2.$$

Thus $D_2 f(\lambda, 0)$ is nonsingular if and only if $\lambda \neq n^2\pi^2$, $n = 1, 2, \ldots$; otherwise

$$N(D_2 f(n^2\pi^2, 0)) = \text{span of } \{\cos n\pi x\}, \quad \text{and}$$

$$R(D_2 f(n^2\pi^2, 0)) = \left\{ u \in C^0 : \int_0^1 u(x) \cos n\pi x \, dx = 0 \right\}.$$

It follows that $\cos n\pi x \notin R(D_2 f(n^2\pi^2, 0))$. Now if we define Z as

$$Z = \left\{ u \in C_0^2 : \int_0^1 u(x) \cos n\pi x \, dx = 0 \right\},$$

then all the hypotheses of the preceding theorem are satisfied. Therefore bifurcation occurs at all points $(n^2\pi^2, 0) \subset \mathbf{R} \times C_0^2$, $n = 1, 2, \ldots$.

EXAMPLE 2. Let Ω be a bounded region in $\mathbf{R}^n$, and consider the problem

$$\Delta u - \lambda\phi(u) = 0 \quad \text{in } \Omega, \qquad au + b\frac{du}{dn} = 0 \quad \text{on } \partial\Omega,$$

where $a^2 + b^2 \neq 0$, and du/dn is the normal derivative of u on $\partial\Omega$. Here ϕ is smooth, $\phi(0) = 0$, and $\phi'(0) \neq 0$. We let $f(\lambda, u) = \Delta u - \lambda\phi(u)$, with the domain and range of f being the obvious analogues of those in Example 1. The linearized operator at $u = 0$ is

$$D_2 f(\lambda, 0)v = \Delta v - \lambda\phi'(0)v, \qquad av + b\frac{dv}{dn} = 0 \quad \text{on } \partial\Omega.$$

If λ_0 is an eigenvalue with one-dimensional null space spanned by u_0 (so, for example, if λ_0 is the principal eigenfunction of the linearized operator; see footnote 3, below), then since the linearized operator is self-adjoint, it follows that $R(D_2 f(\lambda_0, 0))$ is, as in the previous example, the set of v satisfying

$$\int_\Omega v u_0 \, dx = 0.$$

Since $D_1 D_2 f(\lambda_0, 0)v = -\phi'(0)v$, and $\phi'(0) \neq 0$, we see that $D_1 D_2 f(\lambda_0, 0) \notin R(D_2 f(\lambda_0, 0))$. Thus the hypotheses of Theorem 13.5 hold, and $(\lambda_0, 0)$ is a bifurcation point.

In the remainder of this section, we shall describe the very useful "method of Lyapunov–Schmidt." It is a procedure which provides a systematic way to reduce the dimension of the space in which one tries to solve an equation of the form $f(\lambda, u) = 0$ near a singular point. In fact, it often reduces infinite-dimensional problems to finite-dimensional ones.

The idea is to attempt to split the space into two sets and to "project" the equation into each one of them. Of the two equations that are obtained, one is solvable by the implicit function theorem, and the other is often one in a finite-dimensional space, and thus can be attacked by topological techniques, such as degree theory or Morse theory.

Let Λ, B_1, and B_2 be Banach spaces, and suppose $f \in C^k(U, B_2)$ where U is a neighborhood of the origin in $\Lambda \times B_1$. We assume that $f(0, 0) = 0$, and we are interested in solutions of $f(\lambda, x) = 0$ near $(0, 0)$. We make the important assumption that $D_2 f(0, 0)$ is a Fredholm operator (see Chapter 11, §A); i.e.,

(i) $N(D_2 f(0, 0)) \equiv X_1$ is finite dimensional, and
(ii) $R(D_2 f(0, 0)) \equiv X_2$ is a closed subspace of B_2 having finite codimension.

In view of (ii), we may write B_2 as a direct sum $B_2 = Z_2 \oplus X_2$, where Z_2 is a finite-dimensional space. Let Q be the projection operator onto X_2 (i.e., $Q(z, x) = x$). We can also write $B_1 = Z_1 \oplus X_1$. Then if we apply the operators Q and $I - Q$ to the equation $f(\lambda, x) = 0$, we see that

$$f(\lambda, x) = 0 \quad \text{iff } Qf(\lambda, x) = 0 \quad \text{and} \quad (I - Q)f(\lambda, x) = 0.$$

Now the composition Qf maps $\Lambda \times Z_1 \times X_1$ into X_2. If we let $\tilde{X} = \Lambda \times X_1$ and apply the implicit function theorem to $g(\tilde{x}, z_1) \equiv f(\lambda, z_1 + x_1) = 0$, we see that since $D_2 g(0, z_1) = D_2 f(0, z_1)$ is an isomorphism, there is a unique solution $z_1 = \phi(\lambda, x_1)$ near $(0, 0)$, of $Qf = 0$; i.e.,

$$Qf(\lambda, x_1 + \phi(\lambda, x_1)) = 0.$$

Thus, $x_1 + \phi(\lambda, x_1)$ solves $f(\lambda, x) = 0$ iff

$$(I - Q)f(\lambda, x_1 + \phi(\lambda, x_1)) = 0. \tag{13.9}$$

Now $R(I - Q)$ being finite dimensional, means that (13.9) is a *finite* number of equations; it is called the bifurcation equation. If Λ is also a finite-dimensional parameter space, then (13.9) becomes a finite-dimensional system of equations.

EXAMPLE 3. Let $B_1 = \mathbf{R} \times B$, where B is a Banach space, and let S be a compact operator taking B into itself. Suppose further that $r: B_1 \to B$ satisfies (13.5). Let

$$f(\lambda, u) = u - \lambda S(u) + r(\lambda, u),$$

and recall by Theorem 13.2, $T = dS_u$ is a compact linear operator. Now the null space of

$$D_2 f(\lambda_0, 0)(\lambda, \xi) = (I - \lambda_0 T)\xi$$

is the set $\{(\lambda, \xi) : \xi \in N(I - \lambda_0 T)$ and thus has dimension

$$n = 1 + \dim N(I - \lambda_0 T).$$

The standard Riesz–Schauder theory (see [Sr]) implies that $I - \lambda_0 T$ is a Fredholm operator of index zero; thus the codimension of $R(I - \lambda_0 T)$ is also n. The Lyapunov–Schmidt procedure then leads to a system of n equations in $(n + 1)$ unknowns.

§B. Stability of Bifurcating Solutions

In many contexts, a solution $\bar{x}$ of an equation $f(x) = 0$ corresponds to a "steady-state" solution of a dynamical system; i.e., of a time-dependent problem

$$x' = f(x), \qquad = \frac{d}{dt}.$$

A very important problem is to decide whether or not the steady-state solution is stable. Let us recall what is meant by this (cf. Chapter 11, §B). Thus, suppose that $\bar{x}$ is perturbed (slightly!) to the state $\bar{x} + \xi$ and we consider the initial-value problem

$$x' = f(x), \qquad x(0) = \bar{x} + \xi.$$

We want to determine whether this solution tends to $\bar{x}$ as $t \to +\infty$, or even if it stays close to $\bar{x}$ for all $t > 0$. If we consider the "linearized" problem

$$\xi' = df_{\bar{x}}\xi,$$

then if the spectrum of $df_{\bar{x}}$ lies in the left-half plane, the solution ξ decays exponentially to zero as $t \to +\infty$. In this case, it is natural to say that $\bar{x}$ is (*linearly*) *stable*. On the other hand, if the spectrum contains points in the right-half plane, we say that $\bar{x}$ is (*linearly*) *unstable*. We are interested here in the stability properties of bifurcating solutions.[2]

To be a little more specific, let us consider the equations

$$u' = f(\lambda, u), \quad \text{where } f(\lambda, 0) \equiv 0, \quad \lambda \in \mathbf{R}, \tag{13.10}$$

[2] In Chapters 22 and 23 this problem is considered from a generalized "Morse theoretic" point of view.

and suppose that we have the following situation. For $\lambda < \bar{\lambda}$, the spectrum $\sigma(D_2 f(\lambda, 0))$, lies in the left-half plane, Re $z < 0$, while for $\lambda > \bar{\lambda}$, $\sigma(D_2 f(\lambda, 0))$ intersects Re $z > 0$. Then at $\lambda = \bar{\lambda}$, at least one point of the spectrum crosses the imaginary axis; i.e., a point ir, with r real, is in $\sigma(D_2 f(\bar{\lambda}, 0))$. Now if $r = 0$, then, generally speaking, the trivial solution bifurcates into new equilibrium solutions, while if $r \neq 0$ it bifurcates into periodic solutions. This latter circumstance occurs with a so-called "Hopf bifurcation" (see, e.g., [HKW]). In this section we shall be concerned only with bifurcation into equilibria.

We shall consider now two examples which illustrate the theorem we are going to state.

EXAMPLE 1. Let Ω be a bounded domain in $\mathbf{R}^n$ with $\partial\Omega$ smooth and consider the Dirichlet problem

$$\Delta u - \mu u - u^2 = 0 \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega. \tag{13.11}$$

We recall from Chapter 11 §A, that if λ_0 is the principal eigenvalue of Δ, with homogeneous Dirichlet boundary conditions, then $\lambda_0 < 0$, and λ_0 has an associated eigenfunction u_0 which we may assume to be positive on the interior of Ω. Furthermore, it is not too hard to show, using the nodal properties of eigenfunctions, that λ_0 is a simple eigenvalue in the sense that the associated eigenspace is one-dimensional.[3]

Now from Example 2 in the last section $(\lambda_0, 0)$ is a bifurcation point, and the set of solutions near $(\lambda_0, 0)$ consists only of the trivial branch $(\mu, 0)$ and the smooth curve

$$(\mu(s), u(s)) = (\mu(s), su_0 + s\phi(s)), \qquad |s| < \delta, \tag{13.12}$$

where $(\mu(0), u(0)) = (\lambda_0, 0)$. Moreover, for each s with $|s| < \delta$, $\phi(s)$ is in the L_2-orthogonal complement of u_0.

We want to calculate the sign of $\mu'(0)$. To this end, substitute (13.12) into (13.11), divide by s, differentiate with respect to s, and set $s = 0$; this gives

$$\Delta\phi'(0) - \lambda_1\phi'(0) - \mu'(0)u_0 - u_0^2 = 0.$$

Now multiply by u_0 and integrate over Ω to get

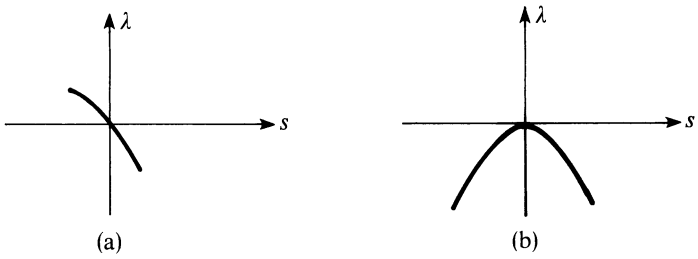
$$\int u_0\Delta\phi'(0) - \lambda_1 \int u_0\phi'(0) - \mu'(0) \int u_0^2 - \int u_0^3 = 0.$$

[3] In fact, the nodes of the nth eigenfunction divide Ω into at most n subdomains; see [CH 1]. If u_2 were an eigenfunction of Δ corresponding to λ_0 and $u_2 \perp u_1$, then u_2 must change sign in Ω. Thus, $(u_2 - u_1)$ would be an eigenfunction corresponding to λ_0 having at least two nodes.

If we integrate the first term by parts, we find

$$\mu'(0)\int u_0^2 = -\int u_0^3,$$

and thus $\mu'(0) < 0$. If we write $\lambda(s) = \mu(s) - \lambda_0$, then this curve is as depicted in Figure 13.3(a); it corresponds to solution branches.



(a) (b)

Figure 13.3

EXAMPLE 2. Consider the Dirichlet problem

$$\Delta u - \lambda u - u^3 = 0 \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega,$$

where Ω is as in Example 1. Here again $(\lambda_0, 0)$ is a bifurcation point, and we have the bifurcating solutions as in (13.11). Computations similar to those in Example 1 show $\lambda'(0) = 0$ and $\lambda''(0) < 0$; this yields Figure 13.3(b).

Definition 13.6. Let X and Y be Banach spaces and let L_0 and $K \in B(X, Y)$. We say that $\mu \in \mathbf{C}$ is a *K-simple eigenvalue* of L_0 (with eigenfunction u_0) if the following three conditions hold:

(i) $\dim N(L_0 - \mu K) = \operatorname{codim} R(L_0 - \mu K) = 1$,
(ii) u_0 spans $N(L_0 - \mu K)$,
(iii) $Ku_0 \notin R(L_0 - \mu K)$.

The terminology comes from the case where $X = Y$, $K = I$, and L_0 is a compact operator. Under these circumstances, $\mu \neq 0$ is an I-simple eigenvalue of L_0 if and only if μ is a simple eigenvalue of L_0. Note too that we could weaken the definition by not requiring L_0 to be a bounded operator. In this case if $X = Y$, $K = I$, and μ is an isolated point in the spectrum of L_0, then μ is an I-simple eigenvalue for L_0 if and only if μ is a simple eigenvalue of L_0.

Let us observe that in this terminology, the hypotheses (i)–(iii) of Theorem (13.5) can be stated as: $L_0 \equiv D_2 f(\lambda_0, 0)$ has 0 as an $L_1 \equiv D_1 D_2 f(\lambda_0, 0)$

simple eigenvalue. The importance of this notion comes from the next lemma, which implies that K-simple eigenvalues "continue" along the bifurcating branches; i.e., for s near zero, $D_2 f(\lambda(s), u(s))$ has a unique (small) K-simple eigenvalue $\mu(s)$. We shall also see that sign $\mu(s) =$ sign $s\lambda'(s)$, and this equation will determine whether the bifurcating solutions are stable or unstable.

Lemma 13.7. *Let μ_0 be a K-simple eigenvalue of L_0, with eigenfunction u_0. Then there exists $\rho > 0$ such that if $\|L - L_0\| < \rho$, L has a unique K-simple eigenvalue $\eta(L)$, with eigenfunction $w(L) = u_0 + z(L)$; here $X = \text{span}\{u_0\} \oplus Z$, and $z(L) \in Z$. Also $\eta(L_0) = \mu_0$, $w(L_0) = u_0$, and the mapping $L \to (\eta(L), w(L))$ is smooth.*

Now let the hypotheses of Theorem 13.5 hold, i.e., as we have seen above, $D_2 f(\lambda_0, 0)$ has 0 as a $D_1 D_2 f(\lambda_0, 0)$ simple eigenvalue. Let $\lambda(s)$ and $u(s) \equiv s(u_0 + \phi(s))$ be the bifurcating curve as provided by this theorem. We shall use the notation provided by this theorem. We shall also use the notation $f'(s) = D_2 f(\lambda(s), u(s))$. Now suppose that $X \subset Y$, the inclusion $i: X \to Y$ is continuous, and 0 is an i-simple eigenvalue of $D_2 f(\lambda_0, 0)$. Then by Lemma 13.7, there exist functions

$$\lambda \to (\gamma(\lambda), v(\lambda)), \qquad s \to (\eta(s), w(s))$$

defined on neighborhoods of λ_0 and 0, respectively, into $\mathbf{R} \times X$, such that $(\gamma(\lambda_0), v(\lambda_0)) = (0, u_0) = (\eta(0), w(0))$, $v(\lambda) - u_0 \in Z$, $w(s) - u_0 \in Z$, and on these neighborhoods,

$$D_2 f(\lambda, 0)v(\lambda) = \gamma(\lambda)v(\lambda), \tag{13.13}$$

$$f'(s)w(s) = \eta(s)w(s). \tag{13.14}$$

Namely, set $\eta(s) = \eta(f'(s))$, $w(s) = w(f'(s))$, where η and w are the functions provided by Lemma 13.7, which is now considered along the curve $(\lambda(s), u(s))$; this gives (13.14). Equation (13.13) comes from the same lemma, now considered along the curve $(\lambda, 0)$; i.e., $\eta(D_2 f(\lambda, 0)) = \gamma(\lambda)$, and $w(D_2 f(\lambda, 0)) = v(\lambda)$. Note that these functions, being compositions of smooth functions, are themselves smooth.

Observe that in the case where $f(\lambda, u) = 0$ is the equilibrium equation of (13.10) then the natural choice of K is the identity I, and equations (13.13) and (13.14) are the linearized equations. Namely, (13.13) is the linearization about $(\lambda, 0)$, and one would study its spectrum as λ crosses λ_0, while (13.14) is the linearization about the bifurcating solution; it would be studied near $s = 0$.

Now the following theorem is used to determine the stability of the bifurcating solution.

Theorem 13.8. *Let the hypotheses of Theorem* 13.5 *hold, and let* γ *and* η *be defined as above. Then* $\gamma'(\lambda_0) \neq 0$, *and if* $\eta(s) \neq 0$ *for* s *near* 0,

$$\lim_{s \to 0} \frac{s\lambda'(s)\gamma'(\lambda_0)}{\eta(s)} = -1. \tag{13.15}$$

We omit the proof of both Lemma 13.7 and Theorem 13.8; see [CRa 1].

We shall show how (13.15) is used to discuss the stability of the bifurcating solutions by returning to Examples 1 and 2 above. First, consider Example 1. Let $(u(s), \mu(s))$ be the bifurcating branch of solutions and as before, let $\lambda(s) = \mu(s) - \lambda_0$. The equation in Example 1 along the bifurcating curve becomes

$$f(\lambda, u) \equiv (\Delta - \lambda - \lambda_0)u - u^2 = 0.$$

Hence $D_2(\lambda, 0)u_0 = (\Delta - \lambda - \lambda_0)u_0 = -\lambda u_0$. Thus using (13.13), we get $-\lambda u_0 = \gamma(\lambda)u_0$, so that $\gamma(\lambda) = -\lambda$. Now we have seen that $\lambda'(0) = \mu'(0) < 0$. Thus, from (13.15) for s near zero,

$$\operatorname{sgn} \eta(s) = -\operatorname{sgn} s.$$

Hence $\eta(s) > 0$ if $s < 0$, and the solution is unstable. For $s > 0$, $\eta(s) < 0$ and the solution is stable. Note too, $\gamma'(\lambda_0) < 0$ so that $\gamma < 0$ if $\mu > \lambda_0$, i.e., $\lambda > 0$ and $(\lambda, 0)$ is stable, while for $\lambda < \lambda_0$, this trivial branch is unstable. These are depicted in Figure 13.4(a). Similarly, for Example 2, we get the situation depicted in Figure 13.4(b). (Of course we are considering the principal eigenvalues in these examples.)

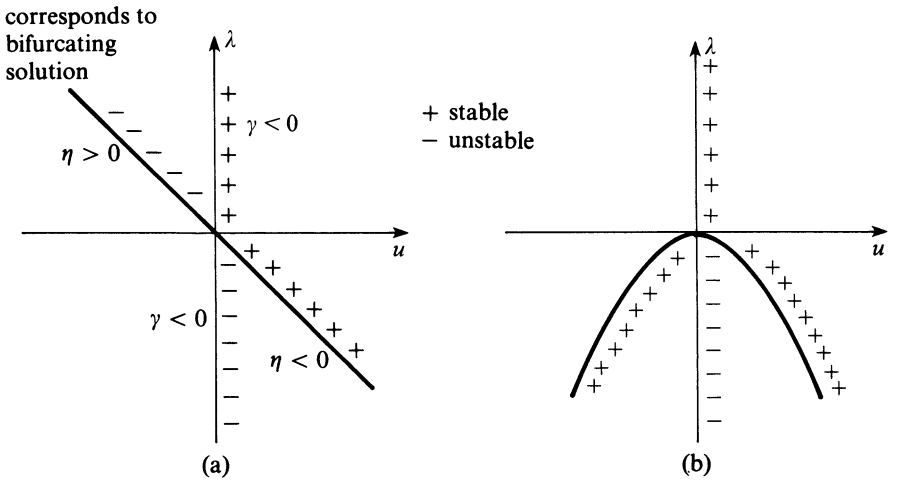


Figure 13.4

§C. Some General Bifurcation Theorems

In the last two sections we studied what may be called "local" bifurcation theory. Our results stemmed from the implicit function theorem, and the bifurcating solutions were only shown to exist near certain distinguished solutions. We shall now give some bifurcation results of a more global nature; the proofs rely on degree theory.

Let B be a Banach space, and let $f \in C(U, B)$ where U is an open subset of $\mathbf{R} \times B$. We assume that f is of the form

$$f(\lambda, u) = u - \lambda Lu + h(\lambda, u), \tag{13.16}$$

where

(i) $1/\mu$ is an eigenvalue of L of odd multiplicity,
(ii) $L : B \to B$ is compact and linear,
(iii) $h : U \to B$ is compact,
(iv) $h(\lambda, u) = o(\|u\|)$ at $u = 0$, uniformly on bounded λ-intervals.

Our first theorem gives conditions for $(\mu, 0)$ to be a bifurcation point. An obvious necessary condition is that $I - \mu L$ is not invertible; hence a necessary condition for $(\mu, 0)$ to be a bifurcation point is that $1/\mu$ is an eigenvalue of L.

Theorem 13.9. *Suppose that* (i)–(iv) *hold. Then* $(\mu, 0)$ *is a bifurcation point of* $f(\lambda, x) = 0$.

Proof. If the result were false, then for $|\lambda - \mu|$ sufficiently small, if $\varepsilon > 0$ is sufficiently small, $d(f(\lambda, \cdot), S_\varepsilon, 0)$ is defined and is independent of λ; here S_ε is the ball of radius ε centered at 0. It follows from Theorem 12.17 that

$$d(f(\lambda, \cdot), S_\varepsilon, 0) = (-1)^{\beta_\lambda},$$

where $\beta_\lambda = \sum \{\text{multiplicity of eigenvalues of } L \text{ greater than } 1/\lambda\}$. If $\lambda_1 > \mu$, and $\lambda_2 < \mu$ then $\beta_{\lambda_1} - \beta_{\lambda_2}$, being the multiplicity of $1/\mu$ (see the remark after the proof of Theorem 12.17), is odd, by hypothesis. This violates the fact that $d(f(\lambda, \cdot), S_\varepsilon, 0)$ is independent of λ, and proves the theorem. □

The second theorem is of a more global nature, in that it is a statement *in the large* about the bifurcation curve which branches out of $(\lambda_0, 0)$.

Theorem 13.10. *Suppose that the above hypotheses* (i)–(iv) *hold with* μ *replaced by* λ_0. *Let* S *denote the closure of the set of nontrivial zeros of* f. *Then* S *contains a component* C (*i.e., a maximal connected subset*) *which meets* $(\lambda_0, 0)$ *and either*

(i) C *is noncompact in* U (*if* $U = \mathbf{R} \times \mathbf{B}$, *this means that* C *is unbounded*), *or*
(ii) C *meets* $u = 0$ *in a point* $(\bar{\lambda}, 0)$ *where* $\bar{\lambda} \neq \lambda_0$ *and* $1/\bar{\lambda}$ *is an eigenvalue of* L.

Proof. Suppose that the theorem is false. Since L is compact, the only possible limit point of the eigenvalues of L is zero (Lemma 11.2). Thus in any finite

λ-interval, there are only a finite number of *characteristic values* (i.e., reciprocals of eigenvalues) of L. Take $\delta > 0$ so small that L has no characteristic values in $I = (\lambda_0 - \delta, \lambda_0 + \delta)$ other than λ_0. Let C be the maximal connected subset of $S \cup \{(\lambda_0, 0)\}$ containing $(\lambda_0, 0)$; our assumption implies that C is compact (at this point it may be that $C = \{(\lambda_0, 0)\}$!). The proof now proceeds in two steps. First we shall find a bounded open subset $\mathcal{O} \subset U$ such that:

(a) $(\lambda_0, 0) \in \mathcal{O}$;
(b) $\partial\mathcal{O} \cap S = \phi$ (∂ taken in U); and
(c) $\mathcal{O}$ contains no nontrivial solutions other than $\{(\lambda, 0): \lambda_0 - \delta < \lambda < \lambda_0 + \delta\}$. Then we shall apply the homotopy invariance of degree in order to get a contradiction.

For step 1, let U_ε be an ε-neighborhood of C in U, where $0 < \varepsilon \ll 1$. Since (ii) holds, and $(\lambda, 0)$ is an isolated solution if λ is not a characteristic value, we may assume that U_ε contains no solution $(\lambda, 0)$ if $|\lambda - \lambda_0| > \varepsilon$. Now let $K = \overline{U}_\varepsilon \cap S$; then K is compact, and by construction, $\partial U_\varepsilon \cap C = \phi$. Set $A = \partial U_\varepsilon \cap S$, and $B = C$; then A and B are closed, nonvoid subsets of K, and $A \cap B = \phi$. At this point we may apply the following lemma [Wh, p. 15]. *Let K be a compact metric space, and let A and B be disjoint, closed, nonvoid subsets of K. Then either there exists a subcontinuum of K which meets both A and B, or there are disjoint closed subsets K_A, K_B containing A and B, respectively, with $K = K_A \cup K_B$.* Since $C \cap \partial U_\varepsilon = \phi$, and C is maximal, the first alternative is excluded, and we conclude that there are disjoint closed subsets K_A and K_B of K with $\partial\overline{U}_\varepsilon \cap S \subset K_A$, $C \subset K_B$, and $K = K_A \cup K_B$. Now let $\mathcal{O}$ be an ε-neighborhood of K_B where $\varepsilon < \operatorname{dist}(K_A, K_B)$; then $\mathcal{O}$ satisfies conditions (a), (b), and (c) above.

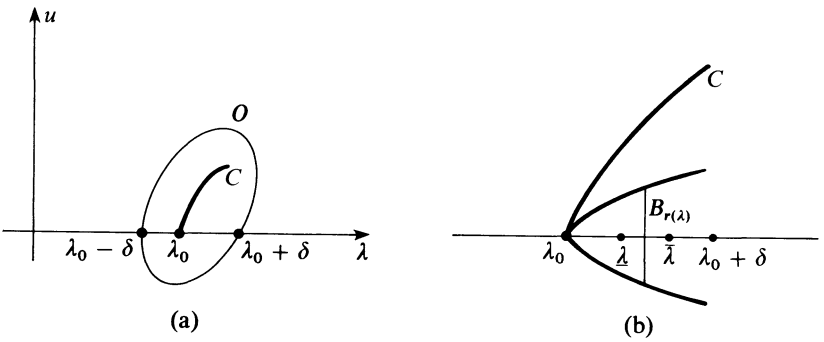


Figure 13.5

Let $O_\mu = \{u \in B: (\mu, u) \in O\}$. For each $\lambda \neq \lambda_0$, choose $r(\lambda) > 0$ such that $(\lambda, 0)$ is the only solution of $f = 0$ on $\{\lambda\} \times B_{r(\lambda)}$, where $B_{r(\lambda)}$ is the ball in B of radius $r(\lambda)$ about $u = 0$ (Figure 13.5(b)).

If $\lambda \neq \lambda_0$, $f = 0$ has no solution in $\{\lambda\} \times \partial(O_\lambda \backslash \overline{B}_{r(\lambda)})$, and thus $d(\lambda) \equiv d(f(\lambda, \cdot), O_\lambda \backslash B_{r(\lambda)}, 0)$ is well defined. Using the homotopy invariance of de-

gree, $d(\lambda)$ is constant for $\lambda > \lambda_0$, $|\lambda - \lambda_0| < \delta$. Since C is compactly contained in O, there is a $\lambda > \lambda_0$ such that $O_\lambda \neq \phi$ and $O_\lambda \backslash \bar{B}_{r(\lambda)}$ contains no solution of $f = 0$. Thus for $\lambda > \lambda_0$

$$d(f(\lambda, \cdot), O_\lambda \backslash \bar{B}_{r(\lambda)}, 0) = 0. \tag{13.17}$$

By a similar argument, this equation holds for $\lambda < \lambda_0$, again for $|\lambda - \lambda_0| < \delta$. Then again using the homotopy invariance, we obtain

$$d(f(\lambda, \cdot), O_\lambda, 0) = \text{const.}, \quad \text{if } |\lambda - \lambda_0| < \delta. \tag{13.18}$$

Now let $\bar{\lambda}$ and $\underline{\lambda}$ be such that $\lambda_0 - \delta < \underline{\lambda} < \lambda_0 < \bar{\lambda} < \lambda_0 + \delta$. Then $O_{\bar{\lambda}} = B_{r(\bar{\lambda})} \cup (O_{\bar{\lambda}} \backslash B_{r(\bar{\lambda})})$, so by the additive property of degree (Chapter 12, §A, Property 5), which is also true in the general Banach space context; see Chapter 12, §B, we have

$$\begin{aligned} d(f(\bar{\lambda}, \cdot), O_{\bar{\lambda}}, 0) &= d(f(\bar{\lambda}, \cdot), O_{\bar{\lambda}} \backslash B_{r(\bar{\lambda})}, 0) + d(f(\bar{\lambda}, \cdot), B_{r(\bar{\lambda})}, 0) \\ &= d(f(\bar{\lambda}, \cdot), B_{r(\bar{\lambda})}, 0), \end{aligned}$$

where we have used (13.17). Hence, from (13.18),

$$d(f(\underline{\lambda}, \cdot), B_{r(\underline{\lambda})}, 0) = d(f(\bar{\lambda}, \cdot), B_{r(\bar{\lambda})}, 0). \tag{13.19}$$

Next, define the homotopy F, taking f to $I\lambda - L$ by

$$F(\lambda, u, \theta) = u - \lambda L u + \theta h(\lambda, u), \qquad 0 \leq \theta \leq 1.$$

If $r(\bar{\lambda})$ is sufficiently small, we see that $F(\bar{\lambda}, u, \theta) \neq 0$ for every $(u, \theta) \in \partial B_{r(\bar{\lambda})} \times [0, 1]$, since (d) holds. Thus we obtain

$$0 \neq d(f(\bar{\lambda}, \cdot), B_{r(\bar{\lambda})}, 0) = d(I - \bar{\lambda} L, B_{r(\bar{\lambda})}, 0),$$

and similarly

$$d(f(\underline{\lambda}, \cdot), B_{r(\underline{\lambda})}, 0) = d(I - \underline{\lambda} L, B_{r(\underline{\lambda})}, 0).$$

These two equations, in conjunction with (13.19), give

$$d(I - \bar{\lambda} L, B_{r(\bar{\lambda})}, 0) = d(I - \underline{\lambda} L, B_{r(\underline{\lambda})}, 0). \tag{13.20}$$

But λ_0 is a characteristic value of L of odd multiplicity; hence by the remark after the proof of Theorem 12.17,

$$d(I - \bar{\lambda} L, B_{r(\bar{\lambda})}, 0) = -d(I - \underline{\lambda} L, B_{r(\underline{\lambda})}, 0) \neq 0.$$

This contradicts (13.20), and the proof is complete. □

We now discuss an application of this theorem.

EXAMPLE. Consider the boundary-value problem

$$f(\lambda, u) \equiv u'' + \lambda\psi(u) = 0, \qquad 0 < x < 1, \tag{13.21}$$

with homogeneous Neumann boundary conditions,

$$u'(0) = u'(1) = 0.$$

Here we assume $\psi(u) = -u(u - a)(u - b)$, where $a < 0 < b$. We have seen, in Example 1 of §A, that the points $(\lambda, u) = (n^2\pi^2, 0)$, $n = 1, 2, \ldots$, are all bifurcation points for $f(\lambda, u) = 0$. We wish to consider the global behavior of these bifurcation branches. (Note that we can write our problem in the equivalent form $u + \lambda K\psi(u) = 0$, (cf. (12.16)), where K is compact, and this equation is easily seen to be of the form (13.16).

To this end, we first recall some standard properties of linear ordinary differential operators (see [CH 1]). Thus, let L be defined on $0 < x < 1$ by $Lv = v'' + a(x)v$, where $\|a\|_\infty < \infty$, with homogeneous Neumann boundary conditions at $x = 0$ and $x = 1$. Then L has a decreasing sequence of simple eigenvalues $\lambda_0 > \lambda_1 > \cdots > \lambda_n > \cdots$, where $\lambda_n \to -\infty$, and if v_n is the eigenfunction corresponding to λ_n, $n \geq 1$, then v_n has precisely n zeros on $(0, 1)$, where at each zero, $v_n' \neq 0$ (i.e., each is a nodal zero).

Now let u be a solution of (13.21) which bifurcates out of $(n^2\pi^2, 0)$; then from Theorem 13.5, we can write $u = u(s)$ where $u(s) = sv_n + s\phi(s)$, for s near zero. Thus for $|s|$ small, we see that u has exactly n-nodal zeros on $(0, 1)$. Since λ_n is a simple eigenvalue of the linearization of (13.21) (so λ_n has *odd* multiplicity equal to 1), we may apply Theorem 13.10. Now a nonzero solution u of (13.21) cannot have $u(\bar{x}) = u'(\bar{x}) = 0$ for some $\bar{x}$ $[0, 1]$, since this would violate the uniqueness theorem for the initial-value problem of the first-order system $u' = w$, $w' = -f(u)$. We conclude that $u(s)$ has exactly n nodal zeros for *all* s for which it is defined. It follows that the branch of bifurcating solutions containing $(\lambda_n, 0)$ cannot come back to the set $u = 0$, and must therefore be unbounded.

In fact, more is true; namely, there cannot be any secondary bifurcation off of this curve; for a proof, see [SW]. The bifurcation diagram takes the form as in Figure 13.6.

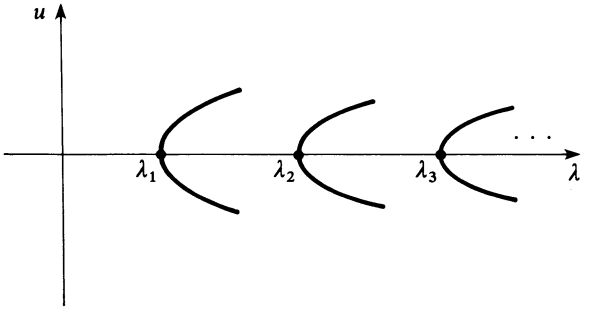


Figure 13.6

§D. Spontaneous Bifurcation; An Example

Up to now, we have considered bifurcation from a particular solution of the equation $f(\lambda, u) = 0$; the distinguished solution usually being one of the form $(\lambda, 0)$. This is what we might term "local" bifurcation. But there is another type of bifurcation which can occur; one of a distinctly global nature. It is what we could call "spontaneous" bifurcation, whereby the solution suddenly "appears," when λ crosses a certain critical value. We shall illustrate this phenomenon by means of a nontrivial important example.

We consider the ordinary differential equation

$$u'' + f(u) = 0, \qquad |x| < L, \tag{13.22}$$

with homogeneous Dirichlet boundary conditions

$$u(\pm L) = 0. \tag{13.23}$$

Here $f(u)$ is the cubic polynomial

$$f(u) = -u(u - a)(u - 1), \qquad 0 < a < \tfrac{1}{2}. \tag{13.24}$$

Notice that $u \equiv 0$ is always a solution; we are interested in nonconstant solutions.

In studying this problem, we shall consider the interval length L as the bifurcation parameter. However, note that if we make the change of variables $y = L^{-1}x$, then (13.22) and (13.23) become

$$L^2 u_{yy} + f(u) = 0, \qquad |y| < 1, \quad u(\pm 1) = 0,$$

so if $\lambda = 1/L^2$, we obtain a problem of the familiar type. We prefer, however, to consider the equation (13.22) because as we shall see presently, its solutions can be given a nice geometric interpretation.

We rewrite (13.22) as a first-order system

$$u' = v, \qquad v' = -f(u), \qquad |x| < L, \tag{13.22$'$}$$

and consider the phase plane for (13.22), as depicted in Figure 13.7. We let $F'(u) = f(u)$, $F(0) = 0$, and observe that the function $H(u, v) = v^2/2 + F(u)$, (the "total energy"), is constant along orbits of (13.22). We define A by $A^2/2 = F(1)$; see Figure 13.7. It is clear that solutions of (13.22)$'$, (13.23) correspond to those orbits of (13.22)$'$ which "begin" on the interval $(0, A)$ on the v-axis (i.e., the line $u = 0$), and "end" on the v-axis, and take "time" (parameter length) $2L$ to make the voyage. We have depicted one such an orbit as the curve joining p to q in Figure 13.7.

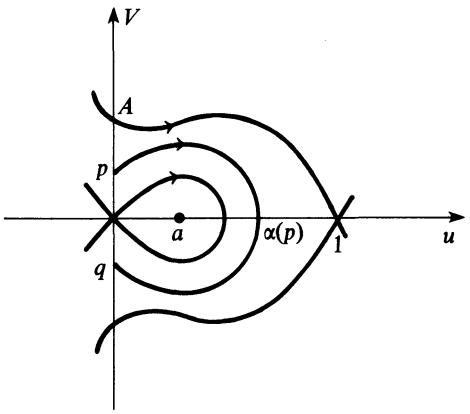


Figure 13.7

Now let $T(p)$ denote the "time" that an orbit starting at p takes to arrive at $\alpha(p)$, as depicted in the figure; here $H(\alpha(p), 0) = H(0, p)$ so that

$$p^2 = 2F(\alpha(p)). \tag{13.25}$$

Using the first equation in (13.22)′, together with the invariance of H along orbits, we find that

$$\sqrt{2}\,T(p) = \int_0^{\alpha(p)} \frac{du}{\sqrt{F(\alpha(p)) - F(u)}}. \tag{13.26}$$

Notice that solutions of (13.22)′, (13.23) correspond to curves for which $T(p) = L$. This leads us to investigate the shape of the graph of T.

To this end, we recall that the domain of T is the open interval $(0, A)$. Furthermore, if p is near 0 or A, then the orbit through p comes near the respective rest points (0, 0) or (1, 0), of (13.22)′, and hence $T(p)$ must be very large. Since T is a smooth function, we see that T must achieve its minimum on $(0, A)$; say at p_0. Obviously $T(p_0) > 0$; see Figure 13.8.

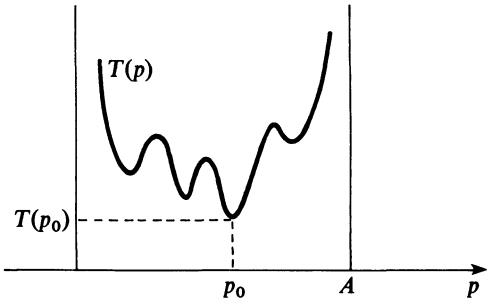


Figure 13.8

Now if $L < T(p_0)$, then it is clear that there are no nonconstant solutions of our problem. But precisely when $L = T(p_0)$, *a nonzero solution suddenly appears*! Viewed from the opposite side, we see that for $L \geq T(p_0)$, there are nonconstant solutions, but as soon as L gets below $T(p_0)$, these solutions disappear; i.e., they "cancel" each other.[4] This cancellation phenomenon will be studied in Chapter 22 in a generalized Morse-theoretic framework; see Theorem 22.33. For now, our interest in this phenomenon is that it is an example of a distinctly different type of bifurcation phenomenon than the one we have hitherto studied.

Actually, we can go much further with this example, and give a precise (qualitative) description of the graph of $T(p)$. By this we mean that we can count the exact number of critical points of T. This will be used in Chapter 24.

Theorem 13.11. *Let f be given by* (13.24) *and let $T(p)$ be defined by* (13.26). *Then T has exactly one critical point.*

Before giving the proof of this result, let us pause to interpret its significance. Thus, the theorem says that the graph of T is *not* as depicted in Figure 13.8, but rather, it is as in Figure 13.9. What this means, is that for

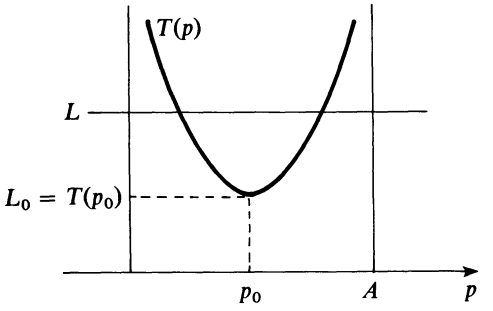


Figure 13.9

$L < L_0, u \equiv 0$ is the only solution; when $L = L_0$, a global bifurcation occurs and we obtain a nonconstant solution, while for every $L > L_0$, there are precisely two nonconstant solutions. (We have already remarked earlier that $u \equiv 0$ is a solution for every $L > 0$.)

If we now view (13.22) as the steady-state equation of the full time-dependent problem

$$u_t = u_{xx} + f(u), \quad |x| < L, \quad t > 0, \qquad u(\pm L, t) = 0, \quad t > 0, \tag{13.27}$$

[4] We may say (*not* tongue in cheek, cf. Chapter 22), that the nonconstant solutions continue to the empty set, while as L increases to $T(p_0)$, a solution appears out of the empty set.

then we have an *exact* count of the number of "rest points" (≡ steady-state solutions) as a function of L. Using the Conley index we can give a global description of *all* solutions of (13.27); this will be carried out in Chapter 24, §D.

Proof of Theorem 13.11. It suffices to prove that T has at most one critical point (since we already know that T has *at least* one critical point). To achieve this, we define $S(\alpha)$ by

$$S(\alpha) = \int_0^\alpha \frac{du}{\sqrt{F(\alpha) - F(u)}}. \tag{13.28}$$

From (13.25) we see that $\alpha'(p) > 0$; thus the equation $\sqrt{2}\,T'(p) = S'(\alpha)\alpha'(p)$, shows that $T'(p) = 0$ if and only if $S'(\alpha) = 0$. Therefore it suffices to prove that S has at most one critical point. For this we will show

$$S''(\alpha) > 0 \quad \text{if } S'(\alpha) = 0. \tag{13.29}$$

To this end we change variables in the integral in (13.28) by writing $u = \alpha \sin \psi$; then

$$S(\alpha) = \int_0^{\pi/2} (F(\alpha) - F(\alpha \sin \psi))^{-1/2} \alpha \cos \psi \, d\psi.$$

This allows us to compute the derivatives; namely,

$$2S'(\alpha) = \int_0^{\pi/2} G^{-3/2}(2G - \alpha G') \cos \psi \, d\psi, \tag{13.30}$$

and

$$2S''(\alpha) = \int_0^{\pi/2} [G^{-3/2}(2G' - \alpha G'' - G') + (2G - \alpha G')(-3/2)G^{-5/2}G'] \times \cos \psi \, d\psi, \tag{13.31}$$

where $G(\alpha) = F(\alpha) - F(\alpha \sin \psi)$.

Now if $S'(\alpha) = 0$, then for this α,

$$2S''(\alpha) = kS'(\alpha) + 2S''(\alpha), \tag{13.32}$$

where k is a constant, to be chosen in a moment. We have from (13.30), (13.31), and (13.32),

$$2S'' = \int_0^{\pi/2} \left[G^{-5/2} \left\{ \frac{-3}{2} G' + kG \right\} (2G - \alpha G') + G^{-3/2}(G' - \alpha G'') \right] \cos \psi \, d\psi.$$

Now put $k = 3/\alpha$ in this equation to get

$$2S'' = \int_0^{\pi/2} \left[G^{-5/2}\left(\frac{3}{2\alpha}\right)(2G - \alpha G')^2 + G^{-3/2}(G' - \alpha G'') \right] \cos\psi \, d\psi$$

$$\geq \int_0^{\pi/2} G^{-3/2}(G' - \alpha G'') \cos\psi \, d\psi. \tag{13.33}$$

But

$$G'(\alpha) = f(\alpha) - f(\alpha \sin\psi) \sin\psi, \tag{13.34}$$

and

$$G''(\alpha) = f'(\alpha) - f'(\alpha \sin\psi) \sin^2\psi.$$

Thus, from (13.33)

$$2S''(\alpha) \geq \int_0^{\pi/2} G^{-3/2}[f(\alpha) - f(\alpha \sin\psi)$$

$$\times \sin\psi - \alpha f'(\alpha) + \alpha f'(\alpha \sin\psi) \sin^2\psi] \cos\psi \, d\psi$$

$$= \frac{1}{\alpha^2} \int_0^{\alpha} (F(\alpha) - F(u))^{-3/2}[\alpha f(\alpha) - uf(u) - \alpha^2 f'(\alpha) + u^2 f'(u)] \, du. \tag{13.35}$$

Writing

$$\theta(x) = 2F(x) - xf(x),$$

gives

$$\theta'(x) = f(x) - xf'(x), \qquad x\theta'(x) = xf(x) - x^2 f'(x),$$

and from (13.35),

$$2S''(\alpha) \geq \frac{1}{\alpha^2} \int_0^{\alpha} (F(\alpha) - F(u))^{-3/2}(\alpha\theta'(\alpha) - u\theta'(u)) \, du. \tag{13.36}$$

Now for the cubic (13.24), we have

$$x\theta'(x) = 2x^3\left(x - \frac{a+1}{2}\right);$$

the graph of this function is depicted in Figure 13.10(a). Thus if $\alpha \geq (a+1)/2$, then (13.36) shows $S''(\alpha) > 0$; that is, $\alpha \geq (a+1)/2$ implies that (13.29) holds. On the other hand from (13.30),

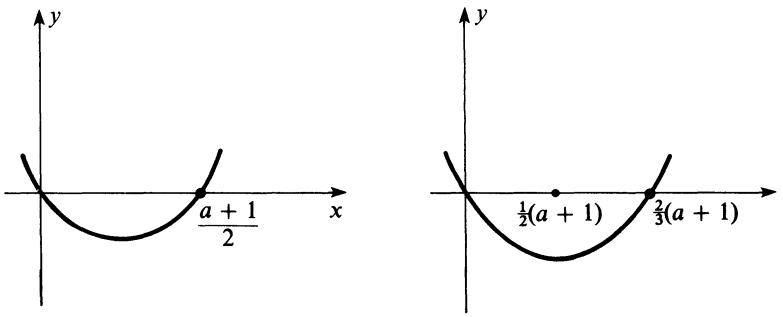
$$2S'(\alpha) = \frac{1}{\alpha}\int_0^\alpha \frac{\theta(\alpha) - \theta(u)}{(F(\alpha) - F(u))^{3/2}}\,du, \tag{13.37}$$

and since

$$\theta(x) = \frac{x^3}{2}[x - \tfrac{2}{3}(a+1)],$$

we see that $\theta(x)$ has the graph as in Figure 13.10(b). Thus $S'(\alpha) \neq 0$ if $\alpha < (a+1)/2$.

We have therefore proved (13.29). From this it follows that S, and thus T has at most one critical point. The proof is complete. □



(a) $y = x\theta'(x)$ (b) $y = \theta(x)$

Figure 13.10

Notes

The proof Theorem 13.3 can be found in Crandall [Cr 2]; see also [Di] and [Ni 2]. Theorem 13.5 is due to C–R [CR 1], following earlier work of Sattinger [Sa 1]. The method of Lyapunov–Schmidt has recently been used by Golubitsky and Schaeffer [GoS] to give a theory of "imperfect bifurcation" via singularity theory. The notion of a K-simple eigenvalue is also due to C–R, as is Lemma 13.7 and Theorem 13.8; see also Sattinger [Sa 1]. Theorem 13.9 due to Krasnoselski [Kr], was the first general bifurcation theorem, and Theorem 13.10 is Rabinowitz' celebrated global bifurcation theorem [Ra]. If one considers positive operators, Dancer [Da] and Turner

[Tu] have shown that one may drop the hypothesis of odd multiplicity in Rabinowitz' theorem. Theorem 13.11 is due to Smoller and Wasserman [SW].

It should here be mentioned again that there is a corresponding theory of bifurcation to periodic solutions, the so-called "Hopf Bifurcation." For a description of this, see the monographs [HKW] and [MM].